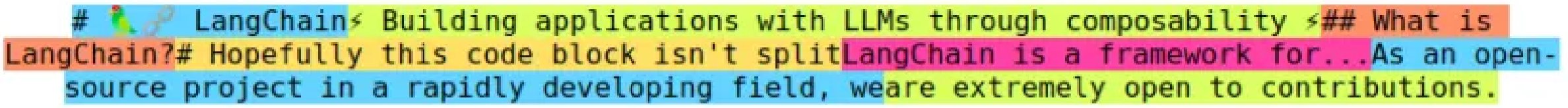
Different Chunking Methods

Bhavishya Pandit

Introduction

Whether you're building the next big AI application, trying to manage mountains of information, or simply wanting to make sense of complex texts, one crucial concept keeps popping up -> **Chunking**.



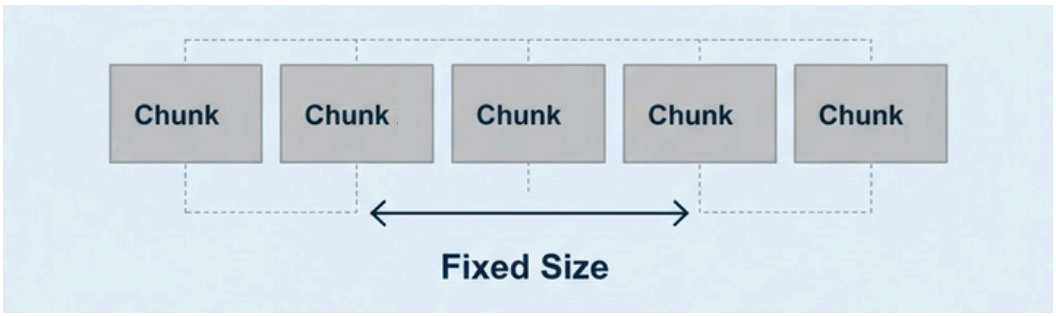
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In this post, we’ll walk through 7 chunking methods that help LLMs in processing information:

- Fixed-Size Chunking
- Content-Defined Chunking
- Delimiter-Based Chunking
- Semantic Chunking
- Recursive Chunking
- Overlapping Chunking
- Paragraph-Based Chunking

Let’s begin with the foundation of it all, fixed -size chunking.

1. Fixed-Size Chunking



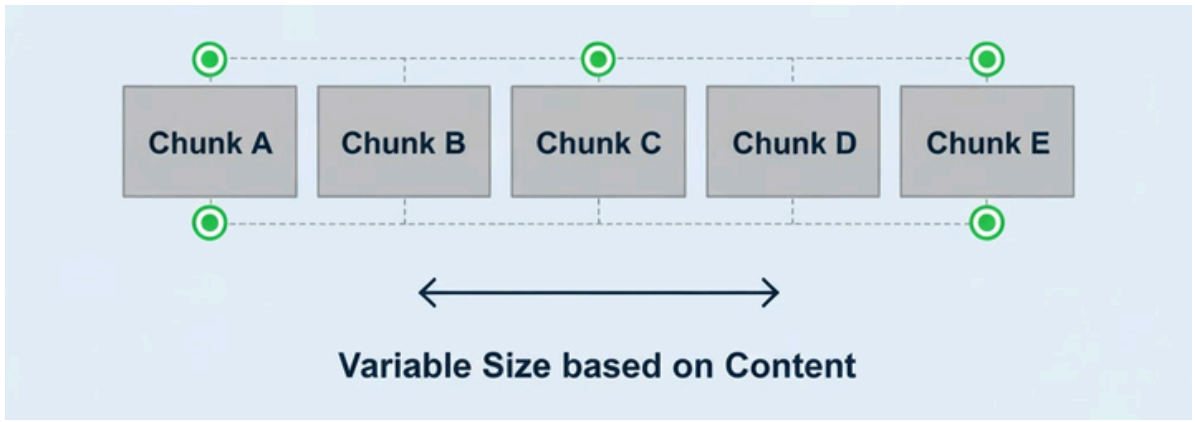
Artificial Intelligence (AI) simulates human intelligence in machines for tasks like visual perception, speech recognition, and language translation. It has evolved from rule-based systems to data-driven models, enhancing performance through machine learning and deep learning



Source

- This is the most basic method.
- You simply decide on a fixed size (e.g., 500 characters, 1000 bytes) and split the data into chunks of that exact size.
- This method's main advantage is its **predictability**.
- Its best use is for handling large binary files (like videos, audio, or large database backups) where semantic meaning is irrelevant.
- It's fast and simple but it pays no attention to the content.

2. Content-Defined Chunking



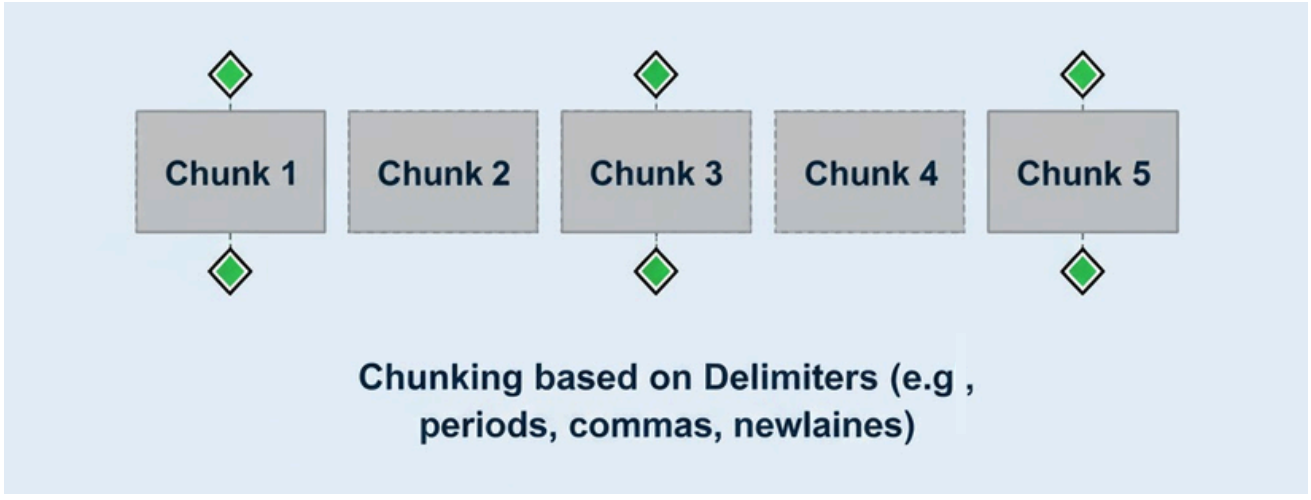
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- It’s a smart chunking method that uses the data's content to find boundaries.
- It slides a window over the data, calculates a hash of that window, and when the hash matches a specific pattern (e.g., ends in 11 zeros), it marks that spot as a chunk boundary.
- Example finding the word ‘STOP’ in a long text and deciding to make a chunk boundary every time you see it. The **chunks will be of variable sizes.**
- Imagine you have a 1GB file, and you change one line in the middle. With Content-Defined Chunking, only the single chunk containing the change is altered.
- Therefore used in **data deduplication and backups.**

3. Delimiter-Based Chunking




```
Artificial intelligence is transforming industries by automating
processes, enhancing decision-making, and providing insights through
data analysis. Machine learning, a subset of AI, enables systems to learn
and improve from experience without explicit programming. Deep learning,
a branch of machine learning, uses neural networks with multiple layers
to model complex patterns in data.
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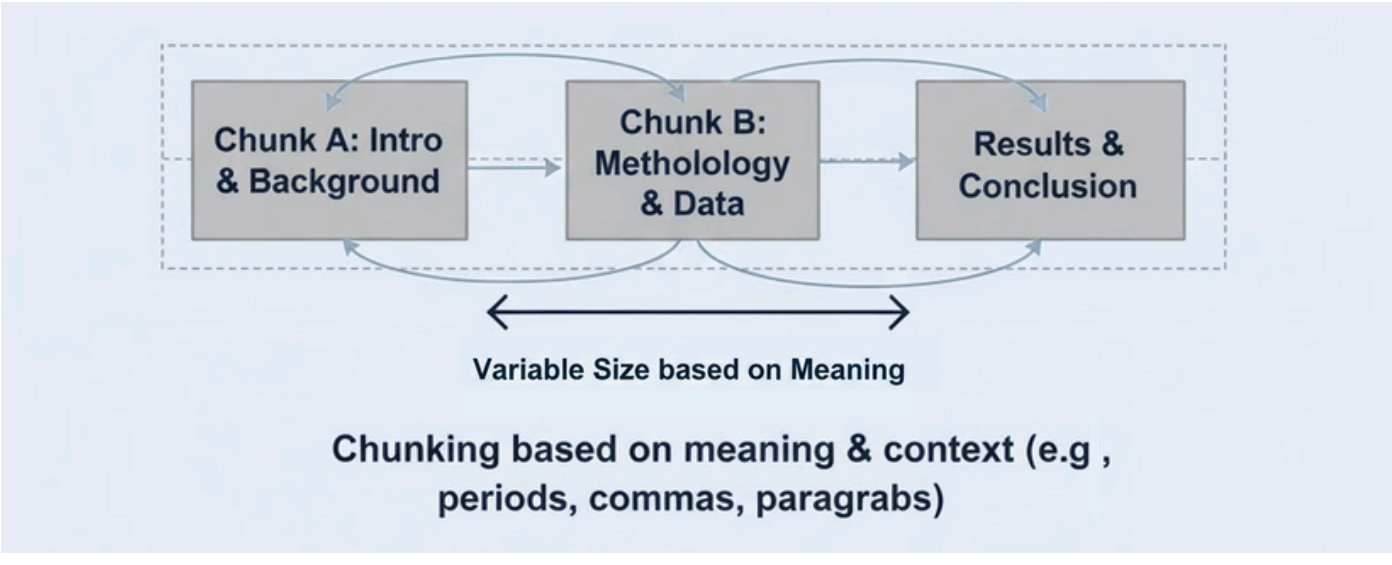
- This method splits data based on a specific character or sequence of characters (called a delimiter), like a full stop.

- It's a very common and intuitive method.

- A very common example is working with a CSV (Comma-Separated Values) files.

- Its very reliable and efficient in processing structured log files like server logs of any application.

4. Semantic Chunking



Artificial Intelligence (AI) simulates human intelligence in machines for tasks like visual perception, speech recognition, and language translation. It has evolved from rule-based systems to data-driven models, enhancing performance through machine learning and deep learning.

Prompt Engineering involves designing input prompts to guide AI models in producing accurate and relevant responses, improving tasks such as text generation and summarization.

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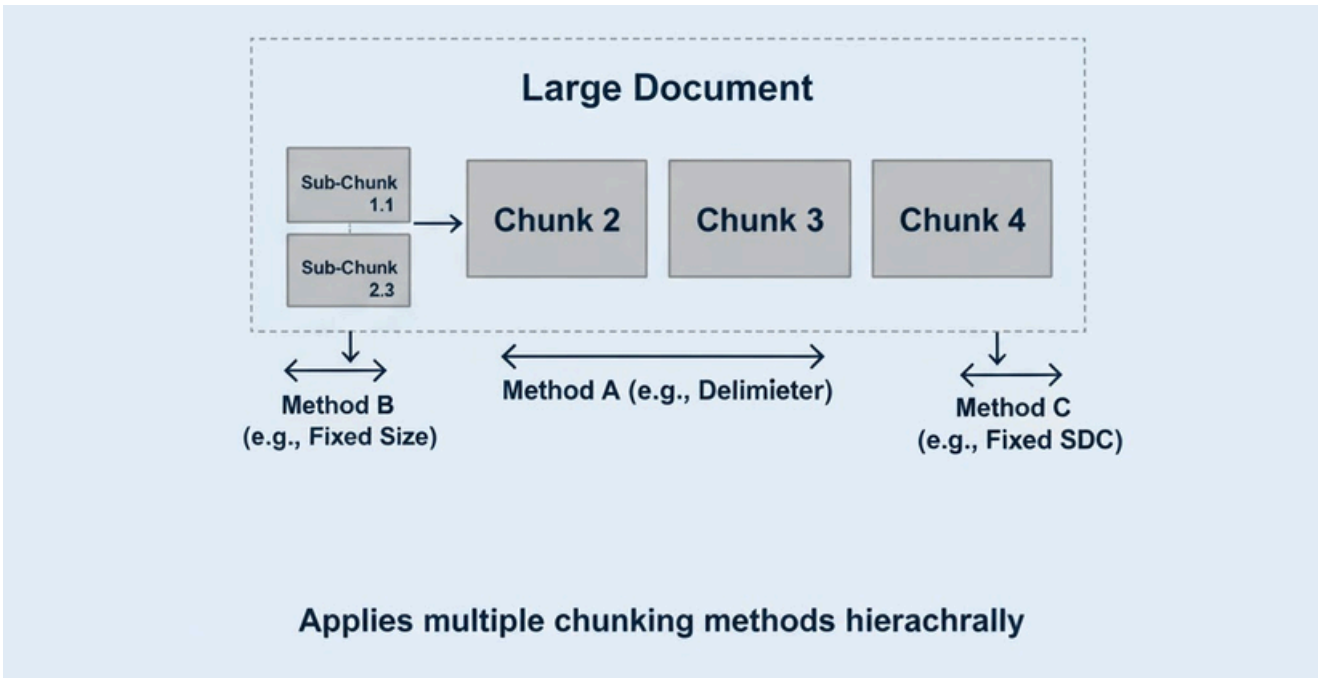
- Semantic chunking is an advanced, AI-powered method that splits text based on meaning or topic, not just characters or punctuation

- It uses language models (like embedding models) to find semantic breaks where the topic of the text shifts.

- For example, given a meeting transcript, semantic chunking wouldn't just split by speaker; it would create chunks related to financial results, actions or marketing campaign.

- This is the state-of-the-art for RAG systems. where the goal of RAG is to find text that answers a user's query.

5. Recursive Chunking



Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.

About Oracle AI Vector Search

Vector Indexes are a new classification of specialized indexes that are designed for Artificial Intelligence (AI) workloads that allow you to query data based on semantics, rather than keywords.

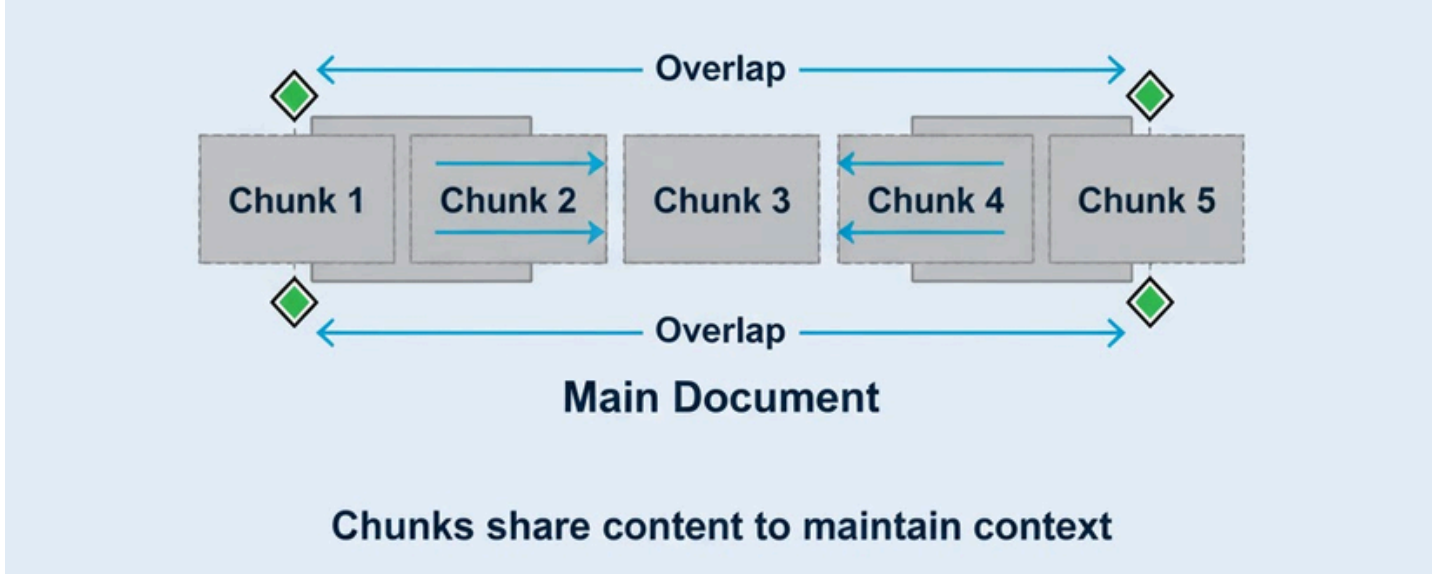
Why Use Oracle AI Vector Search?

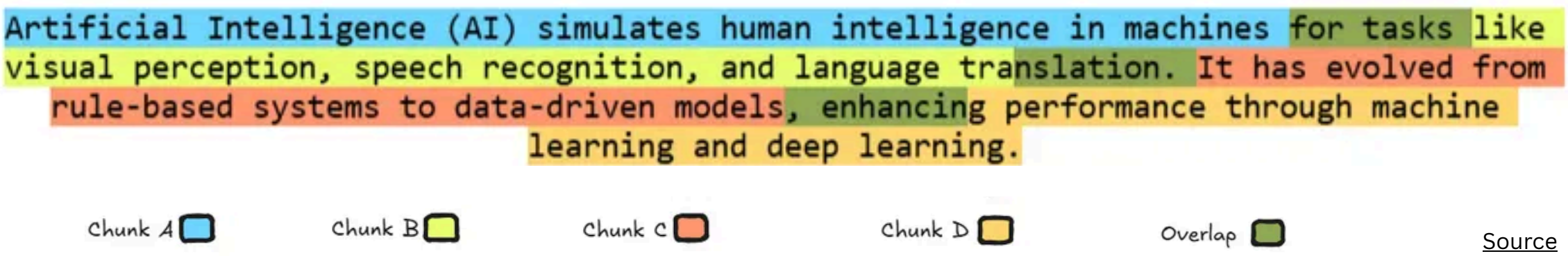
The biggest benefit of Oracle AI Vector Search is that semantic search on unstructured data can be combined with relational search on business data in one single system.

Source

- Its a robust, hierarchical method that tries to split by a list of delimiters, in order.
- It starts with the most semantic delimiter (like paragraphs) and, if any resulting chunk is still too large, it recursively splits that chunk using the next delimiter (like sentences), and so on.
- This is the best all-rounder for general-purpose text processing, especially for data you don't control like the data scrapped from web.

6. Overlapping Chunking

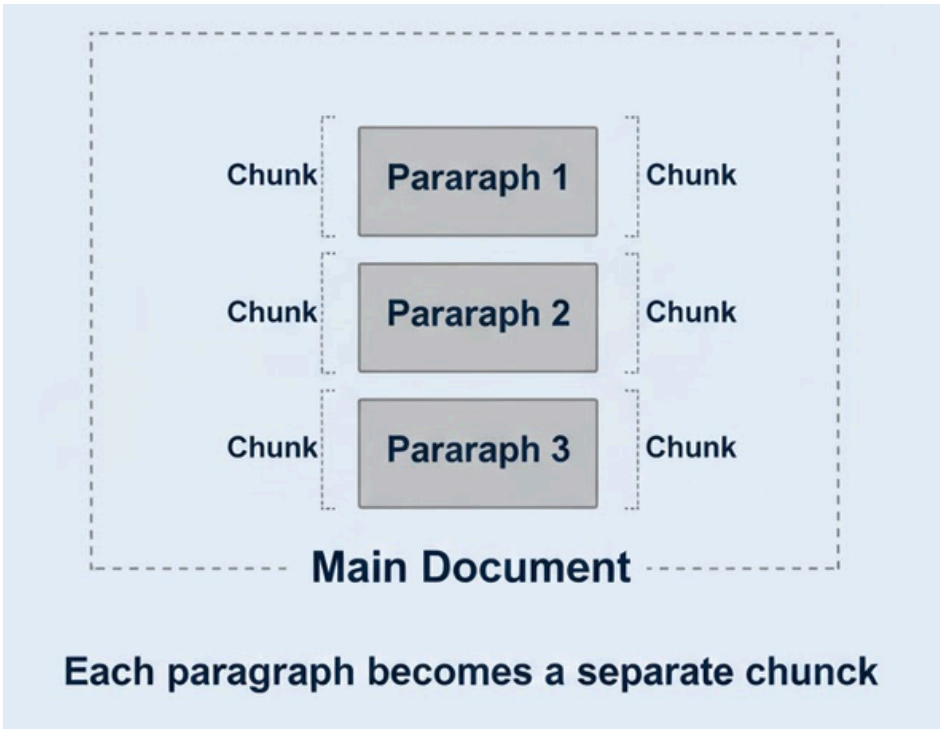


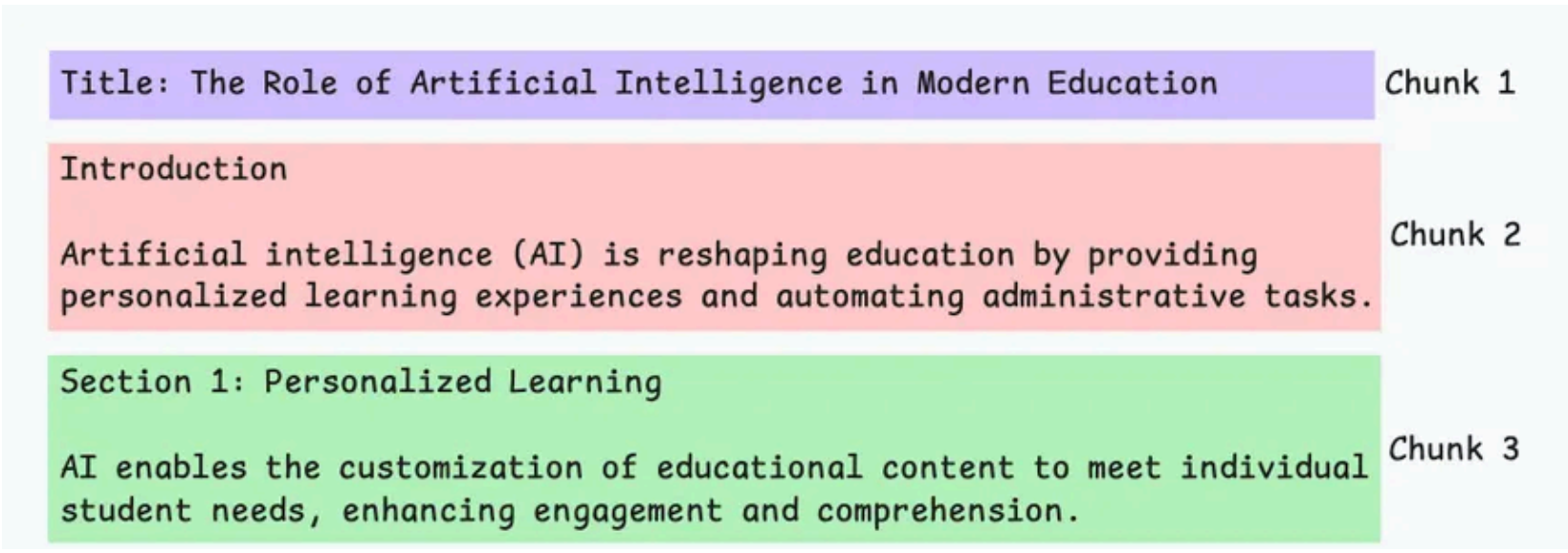


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- This isn't a standalone method but a modifier for other methods
- It ensures that each chunk shares some content with the chunk before and after it.
- This is a simple but effective technique to improve RAG systems, preventing context loss in q&a systems.
- Example: Using Fixed-Size chunking of 100 words with a 30-words overlap.

7. Paragraph-Based Chunking





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- A specific, very common type of delimiter-based chunking where the delimiter is the double-newline character “\n\n”, which typically separates paragraphs.

- It's much better than Fixed-Size and far simpler and faster to implement than full Semantic Chunking, making it an excellent baseline for many NLP tasks.

- This is the go-to method for processing clean, well-formatted text like articles, reports, or legal documents.

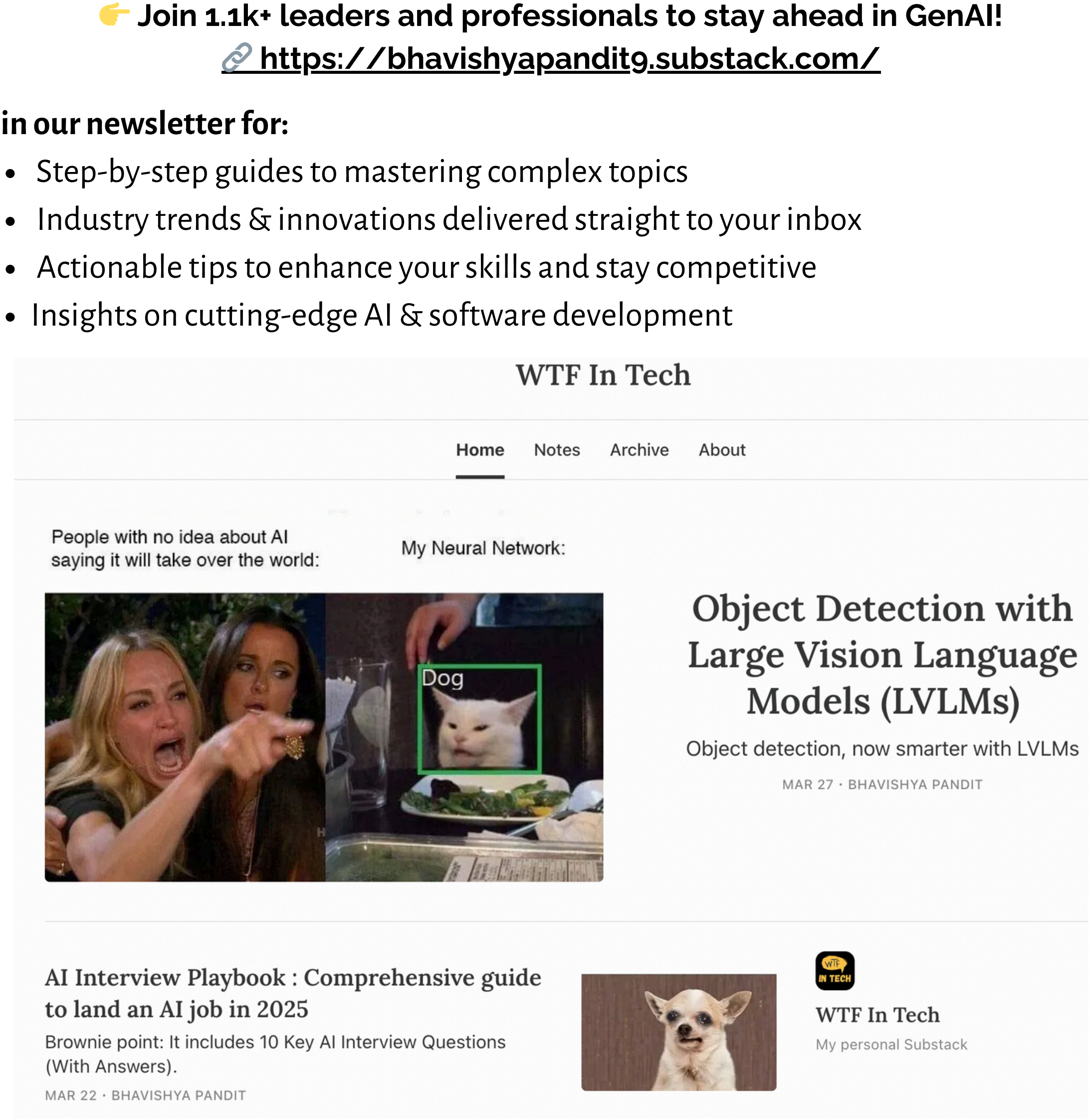
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